E02 Camera Shake

CRE132 worksheet

What you'll make

Camera shake that matches the hit: a small, quick bump when you hit a slime, and a hard half-second shake when a barrel explodes.

There's no code in this sample. You'll use two **Cinemachine** components, an **Impulse Listener** on the camera and an **Impulse Source** on each thing that shakes it, and trigger them from Health's events in the Inspector.

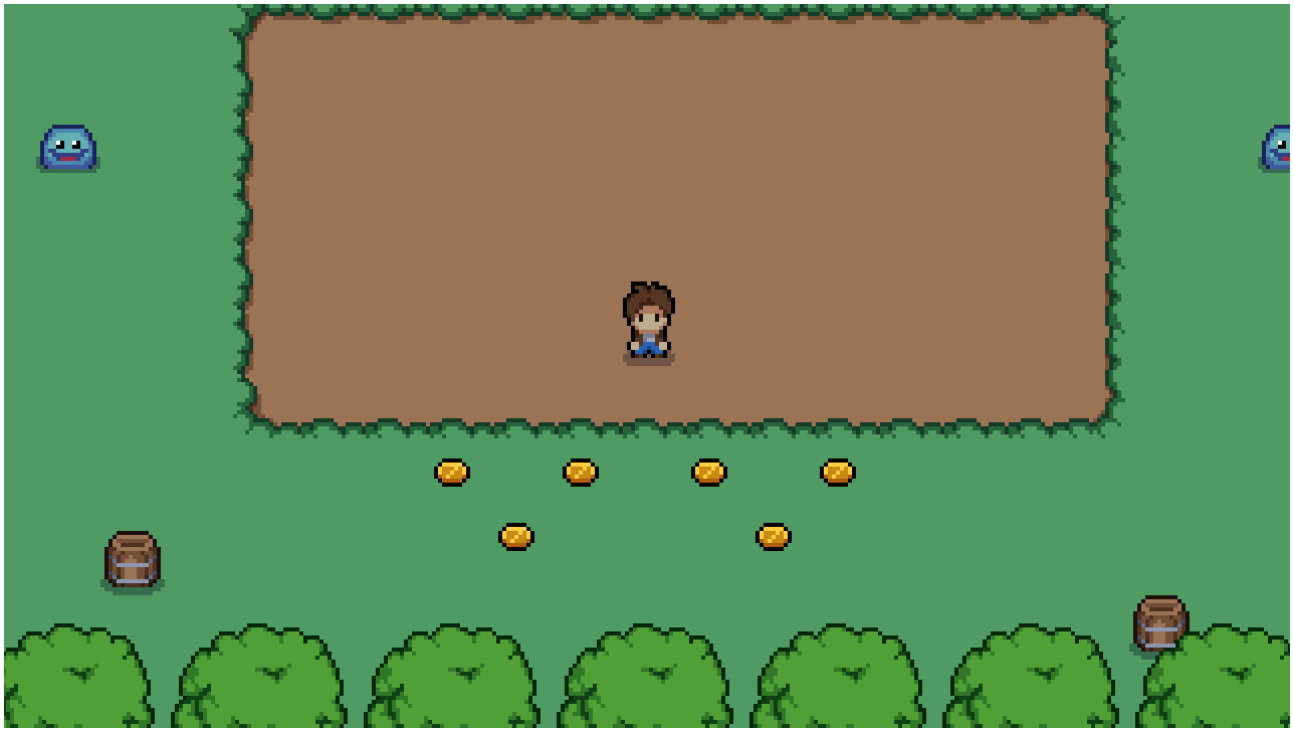


Figure 1: The finished scene: the effects arena with the player, slime dummies and barrels.

Before you start

1. Unzip `CRE132_E02_CameraShake_Starter.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_E02_CameraShake_Starter` folder. Open it with Unity **6000.3.11f1**.
2. Open the scene `Assets/_game/Scenes/Effects/E02_CameraShake`.
3. Press **Play**. Walk with WASD or the arrows, swing the sword with **J** or the left mouse button, and throw a fireball with the right mouse button or **K**. The camera follows you, slimes take hits and barrels explode (E01's **Explosion**), but nothing shakes yet. Press Play again to stop.

The scene has the **Player**, five **Slime Dummies** (inside **Characters**), three **Barrels** and a **CM Follow Camera** that follows the player. Select **CM Follow Camera** and look at its **Cinemachine Camera** component: the **Player** is in **Tracking Target**.

- **A Cinemachine Camera is not a real camera.** It's a set of instructions (follow the player, this zoom). The **Main Camera** has a **Cinemachine Brain**, which copies whatever the Cinemachine Camera says.

- **Shake is added on top.** Cinemachine works out where the camera should be, then the shake nudges it. When the shake ends, the camera is back where it should be.

1. The Impulse Listener

Select **CM Follow Camera** in the Hierarchy. At the bottom of its **Cinemachine Camera** component, open the **Add Extension** menu and choose **CinemachineImpulseListener**. A **Cinemachine Impulse Listener** component appears below. Leave its settings as they are (**Gain** 1).

- **Only cameras with a listener shake.** The listener picks up every impulse in the scene and moves the camera by it. There's nothing to shake it yet, so you'll see the difference in part 2.
- **The listener and the sources are separate**, so any number of things can shake the same camera, and none of them needs to know which camera it is.

2. A small bump: Impulse Sources on the slimes

Expand **Characters** in the Hierarchy. Click the first **Slime Dummy** and Ctrl-click the other four, so all five are selected. In the Inspector, click **Add Component**, then search for **Cinemachine Impulse Source**. With all five still selected, set:

Setting	Value
Impulse Definition > Impulse Shape	Bump
Impulse Definition > Impulse Duration	0.15
Default Velocity	X 0, Y −0.2, Z 0

- **Impulse Shape** is how the shake moves over time: **Bump** is one quick knock, **Explosion** a sharp kick that rattles and dies away, **Recoil** a jolt back, **Rumble** a longer wobble.
- **Default Velocity** is the direction and strength. Y −0.2 knocks the camera slightly down: small, because you'll hit slimes a lot.

Test it before wiring anything: press Play, select one **Slime Dummy**, and in its **Impulse Source** click **Invoke** next to **Test with Force**. The camera bumps. This button only works in Play mode.

3. Shake on a hit: Health's On Damaged

Health already has **UnityEvents**: lists of things to do when it's damaged, healed or dies. Select one **Slime Dummy** and find **On Damaged ()** on its **Health** component. The list is empty. Click **+** to add an entry, then drag that same **Slime Dummy** into the entry's object slot and choose **CinemachineImpulseSource > GenerateImpulse ()** from the function menu. Do the same for the other four slimes.

- **No code:** Health says "I was damaged" and the Impulse Source turns that into a shake. In a script it would be one line, `impulseSource.GenerateImpulse();`.
- **Pick `GenerateImpulse ()` with empty brackets**, which uses the Default Velocity you set. The versions that take a number let you scale it (see **Try it**).

Test it: hit a slime with the sword or a fireball. Every hit gives a small bump.

4. A big shake: the barrels

Expand **Barrels** in the Hierarchy and select all three **Barrels** (click the first, Shift-click the last). Add a **Cinemachine Impulse Source** to them too, with:

Setting	Value
Impulse Definition > Impulse Shape	Explosion
Impulse Definition > Impulse Duration	0.5
Default Velocity	X 0.5, Y −0.6, Z 0

This time use **On Died ()** on each barrel's **Health**. It already calls **ParticleBurst > Play ()** for E01's explosion: click **+** to add a second entry, drag the **Barrel** in and choose **CinemachineImpulseSource > GenerateImpulse ()**. One event can do several things.

Test it: break a barrel. It explodes and the screen shakes hard for half a second, much more than a slime hit.

Check your work

- ☐ **CM Follow Camera** has a **Cinemachine Impulse Listener**; the **Main Camera** doesn't.
- ☐ Every hit on a slime gives a small, quick bump.
- ☐ A breaking barrel shows E01's explosion *and* shakes the screen hard for about half a second.
- ☐ When the shake ends, the camera is back on the player.
- ☐ Compare with `CRE132_E02_CameraShake_Finished` if something behaves differently.

Key points

- **A shake should match the hit:** small for a hit, big for an explosion. Shaking all the time is tiring to watch.
- **Listener and source are separate.** The camera listens; anything with an Impulse Source can shake it.
- **UnityEvents connect components without code.** Health doesn't know about cameras; you wired them together in the Inspector.
- **Some players find shake uncomfortable.** A setting to turn it down is good manners (S06's settings menu could change the listener's **Gain**).

If something goes wrong

- **Nothing shakes at all:** the **CM Follow Camera** has no **CinemachineImpulseListener**, or you added the listener to the **Main Camera** instead.
- **Invoke does nothing:** you're not in Play mode, or the **Default Velocity** is still 0, 0, 0.
- **Hits don't shake but Invoke does:** the **On Damaged** entry has no object in its slot, or its function says **No Function** or **Missing**. Choose **CinemachineImpulseSource > GenerateImpulse ()**.
- **The barrel shakes but doesn't explode any more:** you replaced the **ParticleBurst** entry instead of adding a second one.
- **Only one slime or barrel shakes the camera:** the Impulse Source settings, or the event entry, were set on one object only. Check each one: the Inspector shows a dash (**-**) when the selected objects have different values.

Try it

1. Add a **Recoil** shake when the player throws a fireball. (Hint: an Impulse Source on the **Player**; which script throws the fireball, and does it have an event? If not, call `GenerateImpulse()` from code.)
2. Make the shake stronger the more damage was done: `GenerateImpulseWithForce(damage)` scales the Default Velocity.
3. Try a slime with **Rumble** and a longer **Impulse Duration**. Is it better or worse for a hit? Why?

4. Turn the listener's **Gain** down to 0.3 and up to 3. This is the number a shake-strength slider in a settings menu would change.